# L2 System Modelling - Problem Bank Questions

## L2a. Theoretical Modeling

### Transfer functions

**1.** The linear equation of motion (approximated for small angle about $\theta = 0$) of the single-link pendulum shown in Figure 1 is

$$ml^2\ddot{\theta}(t) + c\dot{\theta}(t) + mgl\theta(t) = u(t)$$

where *m* is the pendulum mass, *c* is the viscous damping, *g* is the gravitational constant, *l* is the distance to the center of mass, and *u* is the applied torque at the rotating base. Find the transfer function $\Theta(s)/U(s)$ of the pendulum assuming the angle starts at zero, $\theta(0^-) = 0$, and is motionless, $\dot{\theta}(0^-) = 0$.

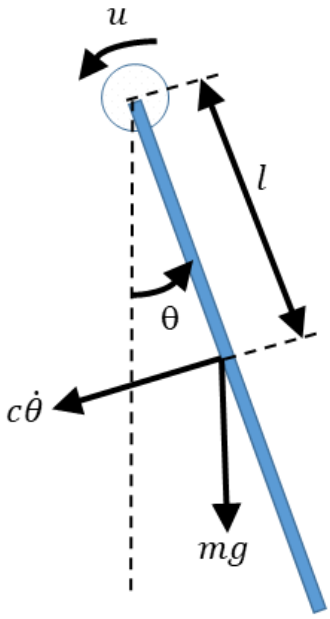
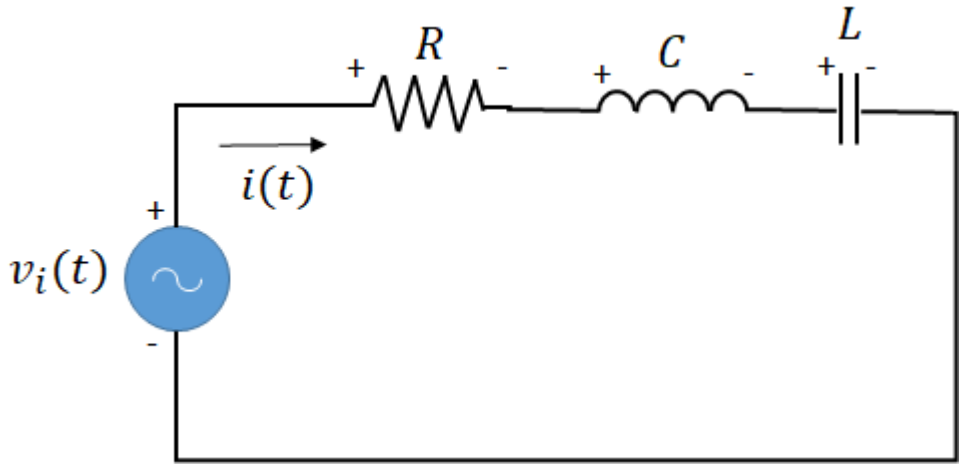



*Figure 1 - Single link pendulum*

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**2.** The RLC circuit shown below has the following equation.

$$v_i(t) = Ri(t) + \frac{1}{C}\int_0^t i\,(\tau)d\tau + L\frac{di(t)}{dt}$$




Find the transfer function of the current, $I(s)$, with respect to the input voltage, $V_i(s)$ with zero initial condition.

**3.** Solve the following differential equation for a unit step input, $u(t)$, and the initial conditions, $y(0^-) = 1$, and is motionless, $\dot{y}(0^-) = 0$. Find $y(t)$.

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5u(t)$$

**Hint**: Use inverse Laplace to obtain $y(t)$.

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## Block Diagrams

**1.** The block diagram in Figure 1 includes two controllers, $C_1(s)$ and $C_2(s)$, to control the plant, $P(s)$. Reduce the block diagram to a single transfer function.

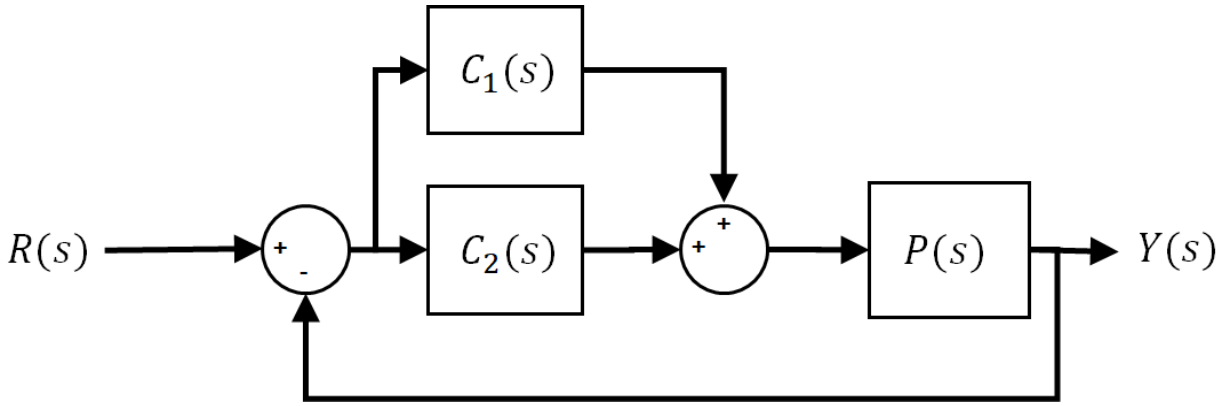



*Figure 1 - Closed-loop system with two controllers*

**2.** The block diagram in Figure 2 uses multiple control loops to stabilize the plant. Reduce the block diagram to a single transfer function.

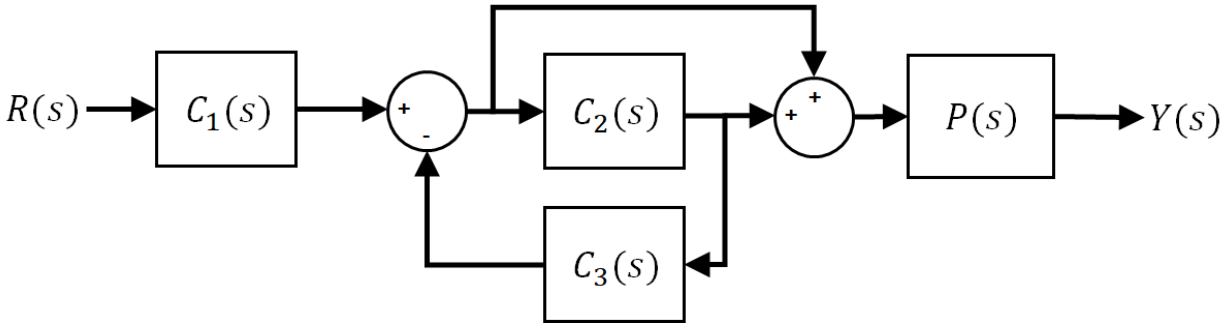



*Figure 2 - Multiple control loops*

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## L2b. State-Space Modelling

**1.** Find the linear state-space matrix representation of the two-loop circuit given below given that the current in each loop is being measured.

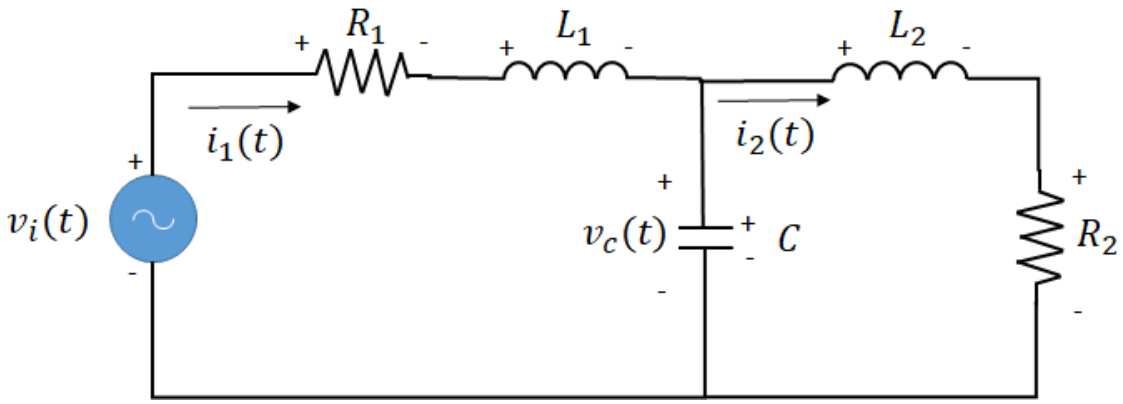



*Figure 1 - Two loop circuit*

The voltage equations for each loop are

$$v_i(t) = R_1 i_1(t) + L_1 \frac{di_1(t)}{dt} + v_c(t)$$

and

$$v_c(t) = L_2 \frac{di_2(t)}{dt} + R_2 i_2(t)$$

The voltage across the capacitor is

$$C \frac{dv_c(t)}{dt} = i_1(t) - i_2(t)$$

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**2.** The diagram shown in Figure 2 shows the free-body diagram of a harmonic drive. The motor produces a torque of $\tau_m$ and has viscous friction coefficient of $b_m$. Although harmonic drives have many advantages (e.g. zero backlash, high torque), their drawback is they can have nonlinear torsional elements. In this case we model the gear train with as a linear spring with a torsional stiffness *k*.

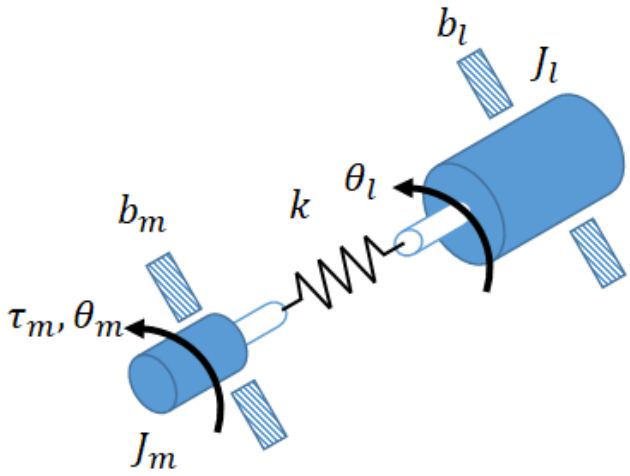



*Figure 2 - Free body diagram of a harmonic drive with torsional flexibilities*

The equation describing the torque input to the motor angle is

$$J_m\ddot{\theta}_m + b_m\dot{\theta}_m + k(\theta_m - \theta_l) = \tau_m$$

The equation for the load angle is

$$J_l\ddot{\theta}_l + b_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$

Only the difference between the load and motor angels is measured directly, i.e. the output is $\theta_l - \theta_m$. Describe both the state and output equations in matrix form.

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**3.** The gantry crane shown in Figure 3 is a prime example of a real-life pendulum-like system. The free-body diagram below shows a force, *u*, being applied to a cart with mass $m_c$. As the cart moves along the support beam, the crane (pendulum) tends to swing back-and-forth. The pendulum has a moment of inertia *J* and the mass $m_p$ and the center of mass from the pivot point is *l*.

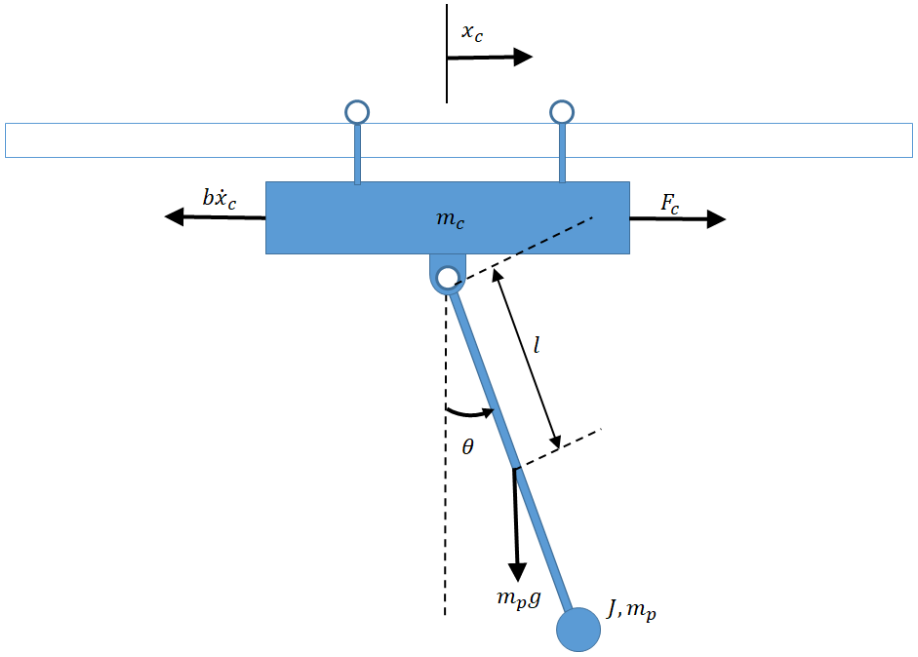



*Figure 3 - Free body diagram of linear gantry carne*

The linearized equations of motion of this system are:

$$m_p l\ddot{x}_c + \left(J + m_p l^2\right)\ddot{\theta} + m_p g l\theta = 0 \qquad \text{Eq. 2}$$

And

$$\left(m_c + m_p\right)\ddot{x}_c + b\dot{x}_c + m_p l\theta = F_c \qquad \text{Eq. 3}$$

Find the linear state-space representation of the system given that the measured variables are the cart position, $x_c$, and the pendulum/crane angular position, θ, and the control input is the force acting on the cart, $F_c$. Include the A, B, C, D matrix form. Assume no sensor noise.

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## L2c. Linearization

**1.** The torsional system below has a nonlinear flexible member with the following equation

$$J\ddot{\theta} + b\dot{\theta} + k\,sin\,\theta\ = \tau$$

where *J* is the moment of inertia, *b* is the damping coefficient, and *k* is the stiffness coefficient. Linearize the system for small angles, $\delta\theta$, about $\theta = 0$ and $\dot{\theta} = 0$, and when the applied input torque is zero, $\tau = 0$.

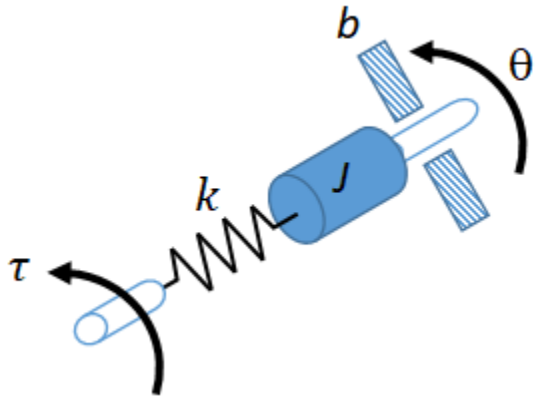



*Figure 1 - Rotary torsional system with a nonlinear flexible coupling.*

**2.** The equation describing the heat transfer via radiation is

$$\dot{T}(t) = T(t)^4 + \epsilon T(t) + u(t)$$

where *T* is the temperature, *u* is the input energy, and ε is a thermal coefficient. Find the linearized equation about the operating point $(T_0, u_0)$ where the temperature does not change, i.e. $\dot{T} = 0$. Express this in terms of the small changes about the operating point, $\delta T = T - T_0$ and $\delta u = u - u_0$ .

**3.** The Ball and Beam device is a classic control problem of stabilizing the ball to a desired position along the beam. The free-body diagram of a Ball and Beam system that uses a rotary servo to control the angle of the beam is illustrated in Figure 2.

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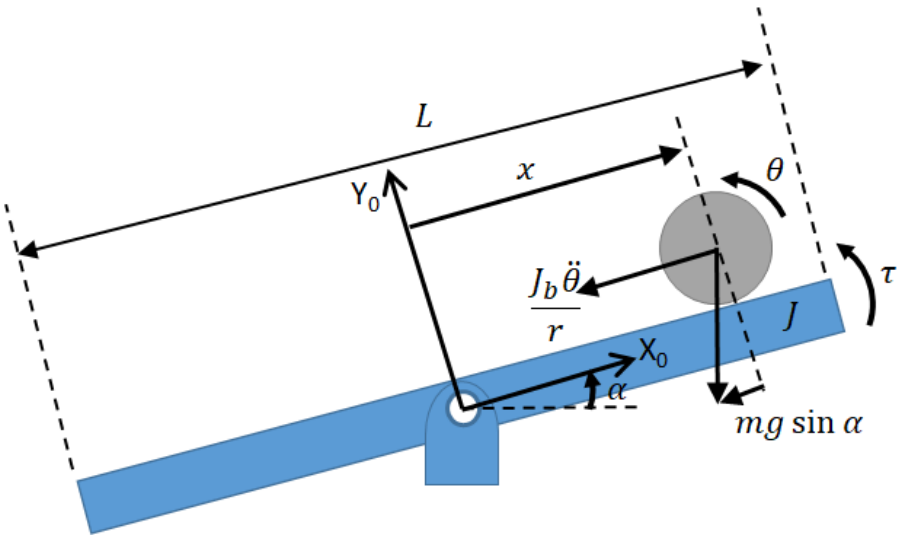



*Figure 2 - Free body diagram of Ball and Beam system*

The nonlinear equations of motions of the system are

$$\frac{7}{5}\ddot{x} + g\ sin\ \alpha\ = 0$$

$$(J + mx^2)\ddot{\alpha} + mg\ x\ cos\ \alpha = \tau$$

where *m* is the mass of the ball, *r* is the radius of the ball, *x* is the linear displacement of the ball along the beam, τ is the applied torque about the beam (the control variable), *L* is the total beam length, α is the angle of the beam relative to the horizontal, and *J* is the moment of inertia of the beam about its rotating axis.

Linearize the system for small angles about the beam horizontal, $\alpha = 0$, and when ball is stabilized in the center of the beam, $x = 0$, and not moving, $\dot{\theta} = 0$.

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## L2d. Experimental Modelling

**1.** The step response of a robot arm driving with a DC motor is shown in Figure 1 .

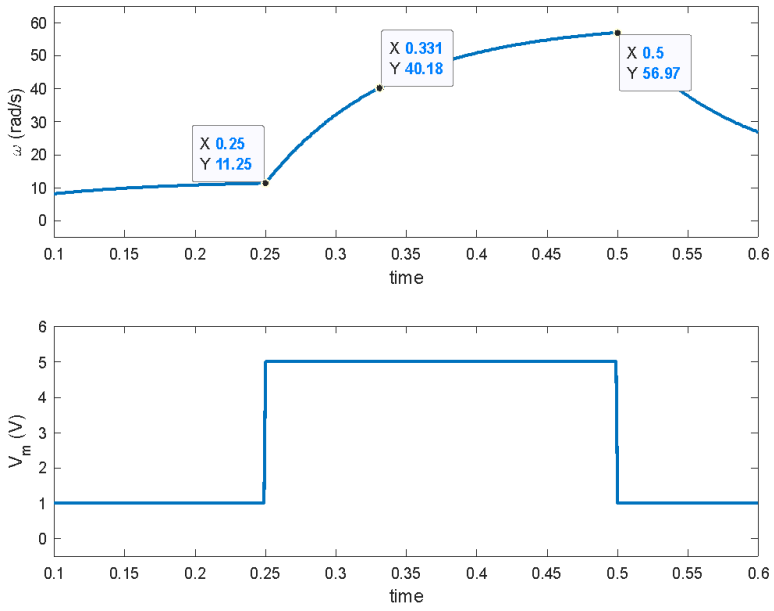



*Figure 1 - Step response of single-link robot arm with DC motor.*

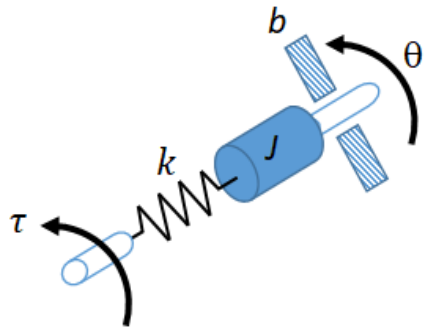
Find the steady-state gain, K, and time constant, τ, parameters of the first-order transfer function

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$$

Use the response data given in Figure 1.

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**2.** Harmonic drives can exhibit flexibilities in their drive. The free-body diagram in is a simplified model of such a system.




The system has an input torque of τ, moment of inertia *J*, a torsional flexibility stiffness of *k*, and a damping coefficient of *b*. The corresponding equation of motion is

$$J\ddot{\theta} + b\dot{\theta} + k\theta = \tau$$

In order to identify the stiffness and damping coefficients, the system is given a small perturbation, i.e. impulse input, and the corresponding angular rate response is measured, as shown in Figure 2.

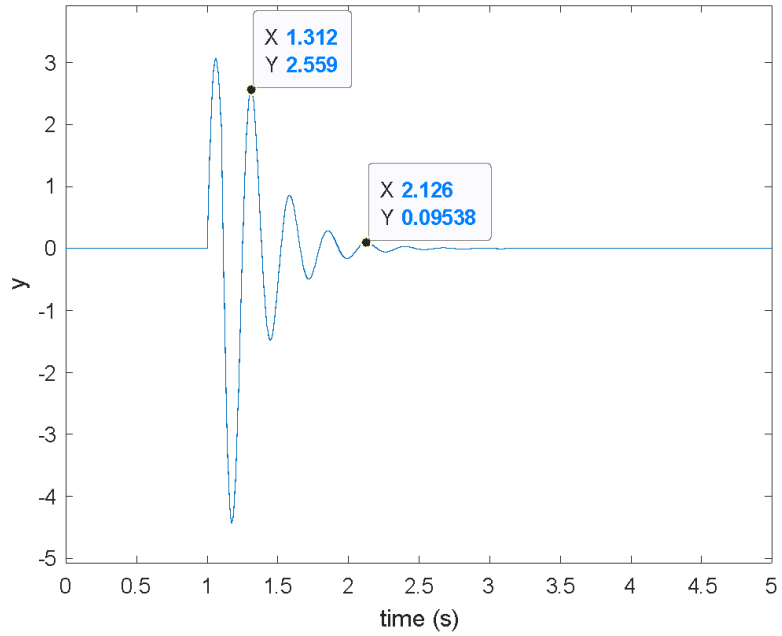



*Figure 2 - Impulse response of harmonic drive*

Find the stiffness, k, and damping, b, coefficients of the system using the experimentally derived response in Figure 2. The moment of inertia of the system is $J = 0.011$ kg-m$^2$.

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## Frequency Response Modelling

**1.** The Bode plot of a DC servo motor is shown in Figure 1. Given that this is a first-order system, find the steady-state gain *K*, cutoff frequency $\omega_c$, and time constant $\tau$.

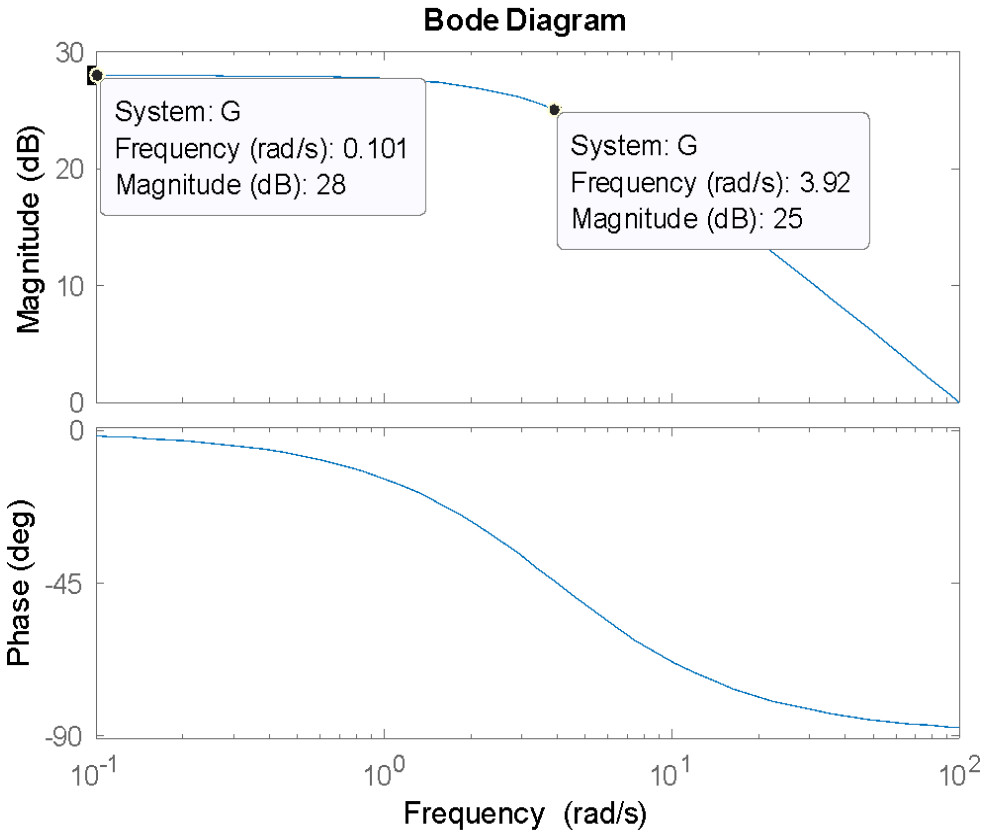



*Figure 1 - Bode plot of DC motor*

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**2.** Find the transfer function of the first-order system shown in the Bode plot in Figure 3.

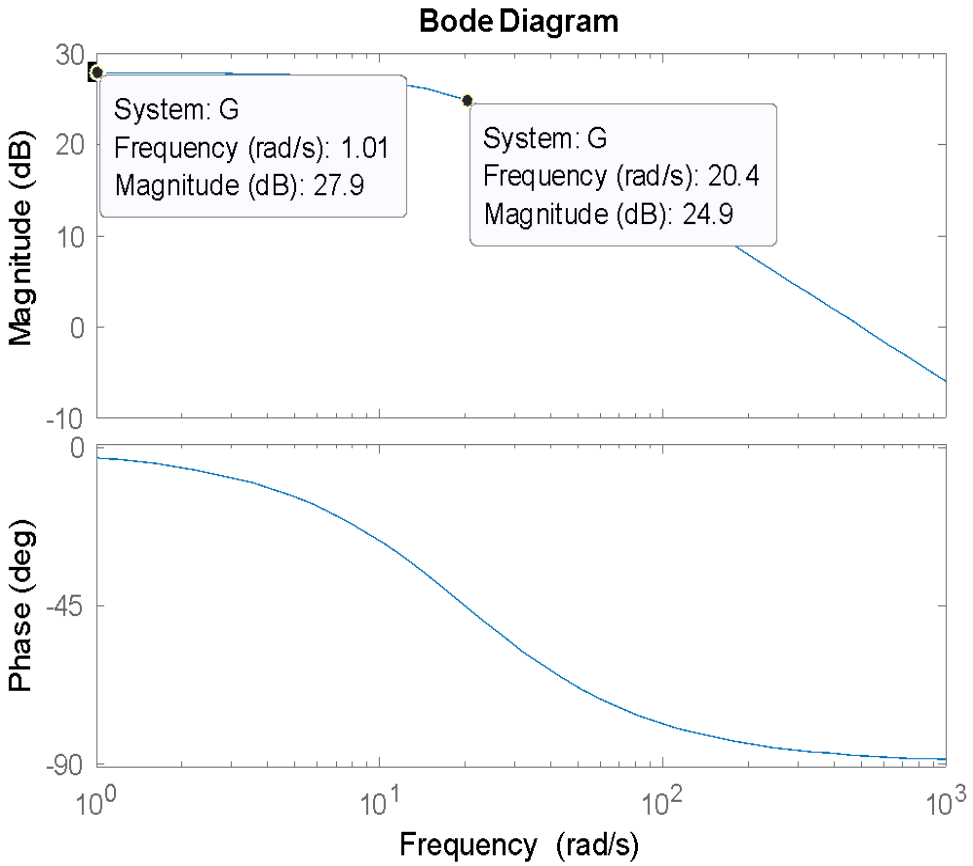



*Figure 3 - Bode plot of first-order system.*

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